

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] DECEMBER 2025 - JANUARY 2026

Paper Code: ECC-207**Subject: Digital Logic and Computer Design****Time: 3 Hours****Maximum Marks: 60****Note: Attempt any five questions including Q.No. 1 which is compulsory. Select one question from each unit. Assume missing data if any.**

Q. No.	Question	Max. Marks	CO(s)
1. (i)	Compare weighted codes with unweighted codes. Give an example of an unweighted code which is self-complementing.	4	CO1
(ii)	Given $PQ' + P'Q = S$, then show that $PS' + P'S = Q$. [Note: Prime notation denotes logical complement / negation].	4	CO1
(iii)	Design a one-bit comparator along with its truth table and proper logic circuit diagram.	4	CO2
(iv)	Differentiate between RISC and CISC architectures in computer organization. What is an interrupt and how does it affect the flow of data?	4	CO4
(v)	Describe the working of a 4-bit Johnson counter with a suitable circuit diagram and state transition table.	4	CO3
UNIT – I			
2. (i)	Implement the given logical expressions with the help of a multiplexer: a) $Y_1 = \Sigma m(0, 2, 3, 8, 12, 14)$ b) $Y_2 = \Sigma m(1, 2, 3, 7, 9, 10, 15)$	6	CO2
(ii)	Design an odd parity generator using minimum number of logic gates.	4	CO2
3. (i)	Implement a full subtractor using a 3×8 decoder and external logic gates.	4	CO2
(ii)	Find the minimal expression for the following function using the Quine–McCluskey (tabular) method: $Y = \Sigma m(1, 2, 5, 7, 9, 12, 13, 15) + \Sigma d(0, 3, 8)$	6	CO1
UNIT – II			
4. (i)	Design a synchronous modulo-10 up/down counter using T flip-flops.	6	CO3
(ii)	Describe the working of a master-slave JK flip-flop with a suitable logic diagram and explain how it eliminates the race-around condition.	4	CO3
5. (i)	Convert a JK flip-flop into a D flip-flop by showing the conversion table and excitation logic.	4	CO3
(ii)	Implement the given Boolean expressions using PAL and PLA: $Y_1 = A'BC' + A'B'C' + AB'C + ABC'$ $Y_2 = A'B'C' + A'B' + AB' + ABC'$ $Y_3 = A'BC' + A'B'C' + ABC + ABC'$	6	CO2

[P.T.O.]

Q. No.	Question	Max. Marks	CO(s)
UNIT – III			
6. (i)	What is the difference between hardwired control and microprogrammed control? Is it possible to have a hardwired control associated with a control memory? Discuss in detail.	5	CO4
(ii)	Explain the different phases of an instruction cycle using a detailed flowchart.	5	CO4
7. (i)	Explain the organization and working of a microprogrammed control unit with a suitable block diagram.	5	CO4
(ii)	What is meant by addressing modes? Describe the different types of addressing modes used in microprogramming and instruction execution in detail with examples.	5	CO4
UNIT – IV			
8. (i)	What is the basic advantage of using interrupt-initiated data transfer over data transfer under program control without an interrupt?	5	CO4
(ii)	How does memory management provide memory protection in modern computer architectures? Explain the underlying hardware mechanisms.	5	CO4
9. (i)	How is interrupt-initiated I/O different from Direct Memory Access (DMA) mode of data transfer? Describe their operational differences in detail.	5	CO4
(ii)	Write comprehensive technical notes on: a) Content-Addressable Memory (CAM) b) Virtual Memory and Address Translation	5	CO4

**** End of Question Paper ****